

Algorithms for Lattice Fermions

User Workshop 2020

5-9th Oct 2020

ALF Tutorial

6th Oct 2020



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ALF basics: *Observables*

6th Oct 2020



Outline

1) Observable categories:

- scalar quantities
- equal-time correlation function
- imaginary-time displaced correlation function

2) Implementation:

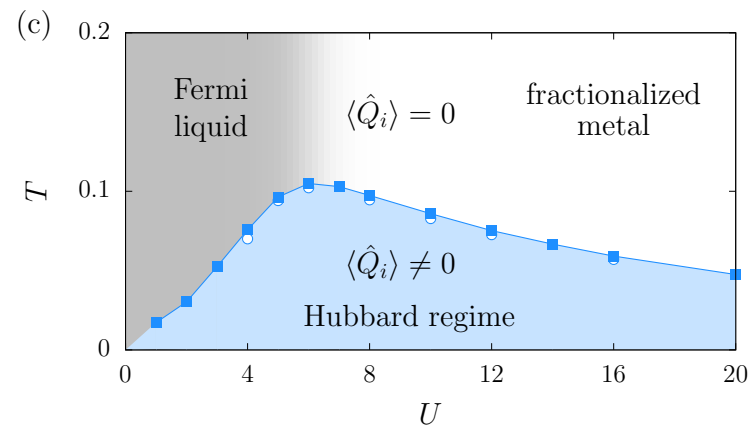
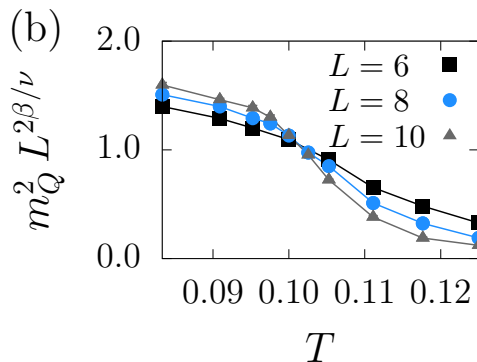
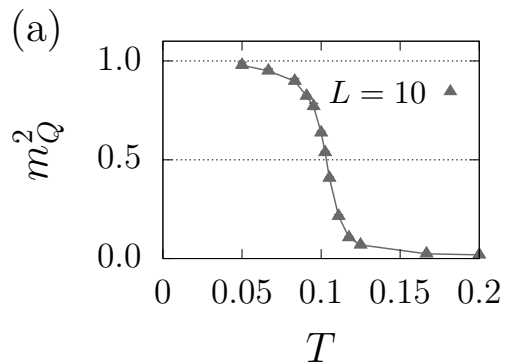
- available run-time settings
- relevant sections in the source code of Hamiltonian module
- predefined plug-and-play routines
- user specified observables

Scalar observable

- Typical examples are the **kinetic, potential, and total energy**, or the **particle number**, as well as
- **Binder cumulants**, which can be used to locate phase transitions, and more.
- Generic form:

$$\hat{O} = \sum_{\vec{i}, x, y} \hat{c}_{\vec{i}, x}^{\dagger} O_{x, y} \hat{c}_{\vec{i}, y}$$

Hohenadler, Assaad; PRL, 2018:



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- Generic form:

$$\hat{O} = \sum_{\vec{i}, x, y} \hat{c}_{i, x}^{\dagger} O_{x, y} \hat{c}_{i, y}$$

Variable	Type	Description	Contribution of configuration C
<code>Obs%N</code>	Int.	Number of measurements	+1
<code>Obs%Ave_sign</code>	Real	Cumulated average sign [Eq. (26)]	$\text{sign}(C)$
<code>Obs%Obs_vec(:)</code>	Comp.	Cumul. vector of observables [Eq. (25)]	$\langle\langle \hat{O}(\cdot) \rangle\rangle_C \frac{e^{-S(C)}}{\text{Re}[e^{-S(C)}]} \text{sign}(C)$
<code>Obs%File_Vec</code>	Char.	Name of output file	

Table 6: Components of a variable of type `Obser_vec` named `Obs`.

Equal-time & time-displaced correlation functions

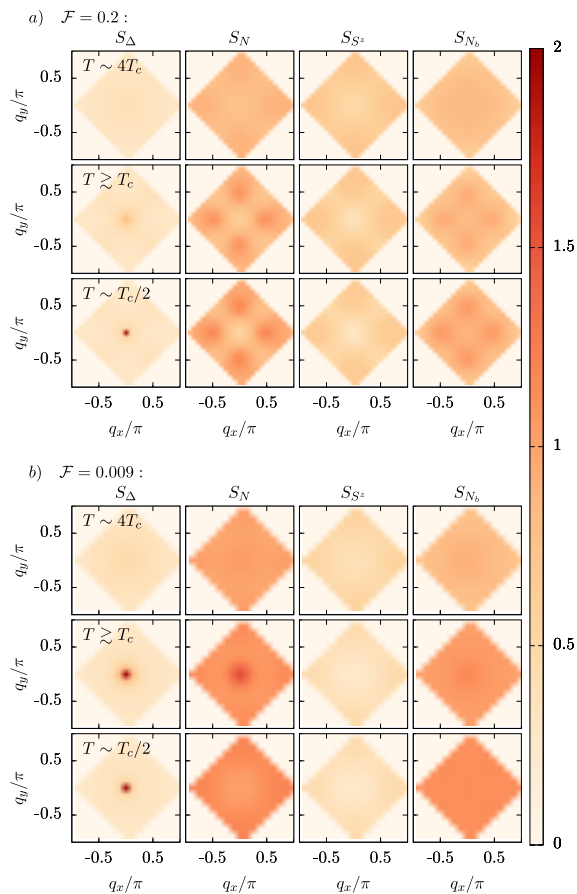
- One is often interested in **spacial resolved** correlation function, e.g., to detect order pattern.

$$S_{\alpha,\beta}(\vec{k}) = \frac{1}{N_{\text{unit-cell}}} \sum_{\vec{i}, \vec{j}} e^{-i\vec{k} \cdot (\vec{i} - \vec{j})} \left(\langle \hat{O}_{\vec{i},\alpha} \hat{O}_{\vec{j},\beta} \rangle - \langle \hat{O}_{\vec{i},\alpha} \rangle \langle \hat{O}_{\vec{j},\beta} \rangle \right)$$

- Typical particle-hole (ph) quantities are the density (CDW) or spin (SDW), but also the equal-time single-particle (p) Green's function
- Imaginary-time** resolution provides access to dynamical properties such as the spectrum.

$$S_{\alpha,\beta}(\vec{k}, \tau) = \frac{1}{N_{\text{unit-cell}}} \sum_{\vec{i}, \vec{j}} e^{-i\vec{k} \cdot (\vec{i} - \vec{j})} \left(\langle \hat{O}_{\vec{i},\alpha}(\tau) \hat{O}_{\vec{j},\beta} \rangle - \langle \hat{O}_{\vec{i},\alpha} \rangle \langle \hat{O}_{\vec{j},\beta} \rangle \right)$$

Hofmann, Chowdhury & Berg;
submitted to PRL:



Equal-time & time-displaced correlation functions

Variable	Type	Description	Contribution of configuration C
<code>Obs%N</code>	Int.	Number of measurements	+1
<code>Obs%Ave_sign</code>	Real	Cumulated sign [Eq. (26)]	$\text{sign}(C)$
<code>Obs%Obs_latt($i - j, \tau, \alpha, \beta$)</code>	Compl.	Cumul. correl. funct. [Eq. (25)]	$\langle \langle \hat{O}_{i,\alpha}(\tau) \hat{O}_{j,\beta} \rangle \rangle_C \frac{e^{-S(C)}}{\text{Re}[e^{-S(C)}]} \text{sign}(C)$
<code>Obs%Obs_latt0(α)</code>	Compl.	Cumul. expect. value [Eq. (25)]	$\langle \langle \hat{O}_{i,\alpha} \rangle \rangle_C \frac{e^{-S(C)}}{\text{Re}[e^{-S(C)}]} \text{sign}(C)$
<code>Obs%File_Latt</code>	Char.	Name of output file	
<code>Obs%Latt</code>	Lattice pointer	Bravais lattice [Tab. 4]	
<code>Obs%Latt_unit</code>	Unit_cell pointer	Unit cell [Tab. 5]	
<code>Obs%dtau</code>	Real	Imaginary time step.	
<code>Obs%Channel</code>	Char.	Channel for Maximum Entropy.	

Table 7: Components of a variable of type `Obser_latt` named `Obs`. Be aware: When creating the observable with the subroutine `Obser_Latt_make`, `Latt` and `Latt_unit` don't get copied but linked, meaning changing them after making the observable still affects the observable.

Equal-time & time-displaced correlation functions

- Equal-time and time-displaced correlation functions are stored using the same structure, only the range of τ differs.

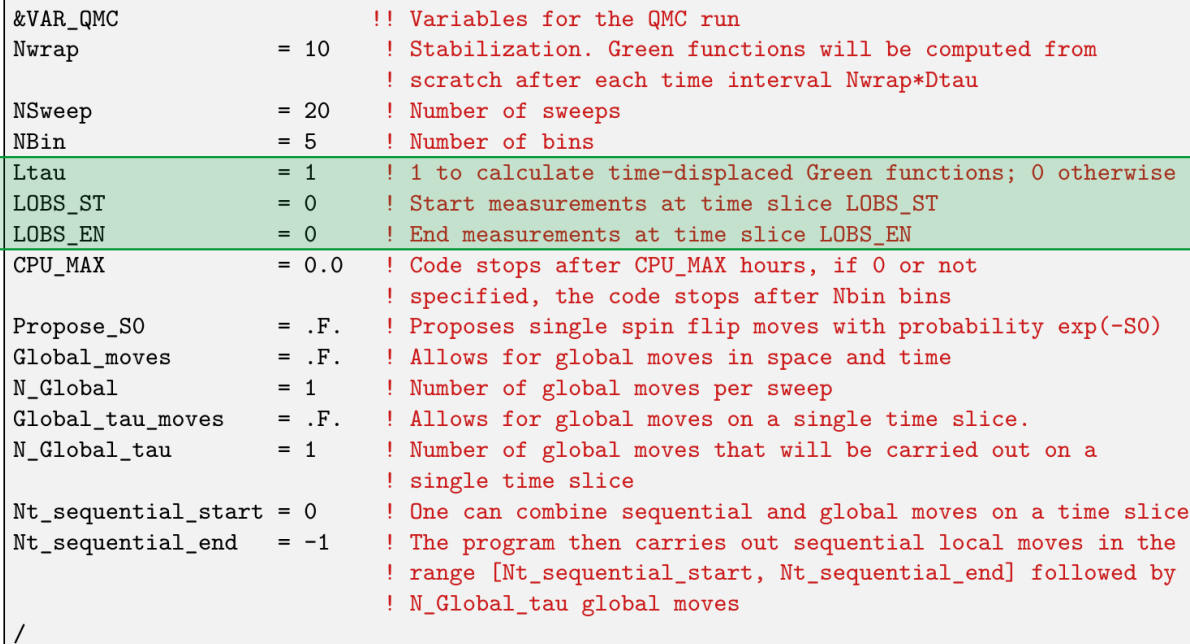
$$S_{\alpha,\beta}(\vec{k}, \tau) = \frac{1}{N_{\text{unit-cell}}} \sum_{\vec{i}, \vec{j}} e^{-i\vec{k} \cdot (\vec{i} - \vec{j})} \left(\langle \hat{O}_{\vec{i},\alpha}(\tau) \hat{O}_{\vec{j},\beta} \rangle - \langle \hat{O}_{\vec{i},\alpha} \rangle \langle \hat{O}_{\vec{j},\beta} \rangle \right)$$

Variable	Type	Description	Contribution of configuration C
Obs%N	Int.	Number of measurements	+1
Obs%Ave_sign	Real	Cumulated sign [Eq. (26)]	sign(C)
Obs%Obs_latt($\mathbf{i} - \mathbf{j}, \tau, \alpha, \beta$)	Compl.	Cumul. correl. funct. [Eq. (25)]	$\langle \langle \hat{O}_{\mathbf{i},\alpha}(\tau) \hat{O}_{\mathbf{j},\beta} \rangle \rangle_C \frac{e^{-S(C)}}{\text{Re}[e^{-S(C)}]} \text{sign}(C)$
Obs%Obs_latt0(α)	Compl.	Cumul. expect. value [Eq. (25)]	$\langle \langle \hat{O}_{\mathbf{i},\alpha} \rangle \rangle_C \frac{e^{-S(C)}}{\text{Re}[e^{-S(C)}]} \text{sign}(C)$
Obs%File_Latt	Char.	Name of output file	

2) Implementation

- **Time-displaced** measurements are expensive both in run-time and data storage. They can be activated by **Ltau = 1**, any other value disabled them.
- **LOBS_ST** and **LOBS_EN** define the measurement interval for **equal-time** observables:

&VAR_QMC		!! Variables for the QMC run
Nwrap	= 10	! Stabilization. Green functions will be computed from scratch after each time interval Nwrap*Dtau
NSweep	= 20	! Number of sweeps
NBin	= 5	! Number of bins
Ltau	= 1	! 1 to calculate time-displaced Green functions; 0 otherwise
LOBS_ST	= 0	! Start measurements at time slice LOBS_ST
LOBS_EN	= 0	! End measurements at time slice LOBS_EN
CPU_MAX	= 0.0	! Code stops after CPU_MAX hours, if 0 or not specified, the code stops after Nbin bins
Propose_S0	= .F.	! Proposes single spin flip moves with probability $\exp(-S_0)$
Global_moves	= .F.	! Allows for global moves in space and time



Data files

- Switch to terminal: `/your/local/path/to/pyALF/Scripts`

Relevant sections in Hamiltonian module (Prog/Hamiltonians/)

- The observable definitions are done in the Hamiltonian module, e.g., *.../ALF/Prog/Hamiltonians/Hamiltonian_Hubbard_mod.F90*
 - **Subroutine** Alloc_obs(Ltau)
Declaration of all observables
 - **Subroutine** Obser(GR,Phase,Ntau)
equal-time measurements
 - **Subroutine** ObserT(NT, GT0, G0T, G00, GTT, PHASE)
time-displaced measurements
 - **Subroutine** Init_obs(Ltau) (exported to Hamiltonian_Hubbard_include.h)
 - **Subroutine** Pr_obs(Ltau) (exported to Hamiltonian_Hubbard_include.h)

- ALF offers an ever-growing list of predefined observable:
 - **Subroutine** Predefined_Obs_eq_Green_measure
single-particle Greens function
 - **Subroutine** Predefined_Obs_eq_Den_measure
density correlation function
 - **Subroutine** Predefined_Obs_eq_SpinSUN_measure
spin correlation function for SU(N) symmetric models
 - **Subroutine** Predefined_Obs_eq_SpinMz_measure
spin correlation function for Mz symmetric model
 - For time-displace versions, replace Obs_eq by Obs_tau
 - Something missing?
Feel free to contribute by opening an issue on GitLab or even implement it yourself and share your solution with others through GitLab!

User specified observable

- Wick's theorem can be applied for a **fixed field configuration**.
- **Generic N-point correlation** function are determined from the **single particle Greens function**:

$$\sum_{\sigma, \sigma'} \langle \langle \hat{c}_{i_1, \sigma}^\dagger \hat{c}_{i_2, \sigma} \hat{c}_{j_1, \sigma'}^\dagger \hat{c}_{j_2, \sigma'} \rangle \rangle = \sum_{\sigma, \sigma'} \left(\langle \langle \hat{c}_{i_1, \sigma}^\dagger \hat{c}_{i_2, \sigma} \rangle \rangle \langle \langle \hat{c}_{j_1, \sigma'}^\dagger \hat{c}_{j_2, \sigma'} \rangle \rangle + \langle \langle \hat{c}_{i_1, \sigma}^\dagger \hat{c}_{j_2, \sigma'} \rangle \rangle \langle \langle \hat{c}_{i_2, \sigma} \hat{c}_{j_1, \sigma'}^\dagger \rangle \rangle \right)$$

- **Only two imaginary times** τ and $\tau = 0$ may occur in time-displaced correlation function

$$\text{GR}(\mathbf{x}, \mathbf{y}, \sigma) = \langle \hat{c}_{x, \sigma} \hat{c}_{y, \sigma}^\dagger \rangle, \quad (127)$$

$$\text{GRC}(\mathbf{x}, \mathbf{y}, s) = \delta_{x, y} - \text{GR}(\mathbf{y}, \mathbf{x}, s) \quad (130)$$

$$\langle \langle \hat{c}_{x, s}^\dagger \hat{c}_{y, s} \rangle \rangle$$

$$\begin{aligned} \text{GT0}(\mathbf{x}, \mathbf{y}, s) &= \langle \langle \hat{c}_{x, s}(Nt\Delta\tau) \hat{c}_{y, s}^\dagger(0) \rangle \rangle = \langle \langle \mathcal{T} \hat{c}_{x, s}(Nt\Delta\tau) \hat{c}_{y, s}^\dagger(0) \rangle \rangle \\ \text{G0T}(\mathbf{x}, \mathbf{y}, s) &= -\langle \langle \hat{c}_{y, s}^\dagger(Nt\Delta\tau) \hat{c}_{x, s}(0) \rangle \rangle = \langle \langle \mathcal{T} \hat{c}_{x, s}(0) \hat{c}_{y, s}^\dagger(Nt\Delta\tau) \rangle \rangle \\ \text{G00}(\mathbf{x}, \mathbf{y}, s) &= \langle \langle \hat{c}_{x, s}(0) \hat{c}_{y, s}^\dagger(0) \rangle \rangle \\ \text{GTT}(\mathbf{x}, \mathbf{y}, s) &= \langle \langle \hat{c}_{x, s}(Nt\Delta\tau) \hat{c}_{y, s}^\dagger(Nt\Delta\tau) \rangle \rangle. \end{aligned} \quad (131)$$